COURIER MANAGEMENT SYSTEM

Salesforce-Based Project Documentation Project Documentation
Prepared for Project Submission

Student's Name: U. Punithavalli,
R. Kanimozhi
D.vijayashri

Register Number: 24131241802522026,
24131241802522013,
24131241802522038

Department: B.Sc. Computer Science

College: Government Arts & Science College - Vadalur

Academic Year: 2026–2027

College Code: ANM-124

University: Annamalai University

Document Note:

This document provides the main steps required for the Courier Management System using Salesforce.

It covers the important areas such as creating objects and fields, setting up users and profiles, managing courier details, creating reports and dashboards, and configuring the required Salesforce settings.

Only the necessary information and steps are included to make the document simple, clear, and easy to understand.

Contents:

1. Project Overview
2. Problem Statement
3. Objectives
4. Requirements and Technology
5. System Design
6. Salesforce Setup
7. Create Custom Objects
8. Create Fields
9. Create Relationships
10. Create Tabs and Lightning App
11. Validation Rules
12. Page Layouts
13. Automation Using Flow
14. Profiles and Permissions
15. Roles and Users
16. Reports and Dashboard
17. Testing
18. Advantages and Limitations
19. Future Scope
20. Conclusion

1. Project Overview

The Courier Management System (CMS) is a Salesforce-based application for managing courier bookings, shipment information, delivery agents, branches, delivery updates, invoices, reports, and dashboards in one place.

The system replaces scattered manual processes with a centralized cloud platform. Staff can create and update shipment records, assign delivery agents, monitor status, manage billing information, and view operational reports.

Main Modules:

Customer Management

Shipment Management

Delivery Agent Management

Branch Management

Delivery Updates

Invoice Management

Reports and

Dashboard s User

Access

and

Security

2.Problem Statement:

Courier operations can become difficult to manage when shipment details are maintained manually or in separate files.

Tracking updates may be delayed, delivery agents may be assigned manually, and managers may not have a clear view of operational performance.

The proposed CMS provides a centralized Salesforce solution to store shipment data, automate status updates, control user access, and generate reports.

3.Objectives

- Store courier information in a centralized Salesforce system.
- Create and track shipments from booking to delivery.
- Maintain customer, branch, agent, and invoice information.
- Reduce incorrect data using validation rules.
- Automate shipment status updates using Flow.
- Provide role-based access to project users.
- Generate reports and dashboards for management.

4. Requirements and Technology

Functional Requirements

No.	Requirement
FR-1	Manage customer information
FR-2	Create and track shipments
FR-3	Assign delivery agents
FR-4	Record delivery status updates
FR-5	Manage invoices and payment status
FR-6	Generate reports and dashboards

Non-Functional Requirements

Area	Requirement
Usability	Simple Salesforce Lightning interface
Security	Role and profile based access
Reliability	Validation and automated processing
Performance	Quick record and status updates
Availability	Cloud-based access
Scalability	Can support additional branches and records

Technology Stack

Layer	Technology
Platform	Salesforce
User Interface	Salesforce Lightning
Data	Salesforce Custom Objects
Automation	Salesforce Flow
Validation	Validation Rules
Security	Profiles, Roles and Permissions
Analytics	Salesforce Reports and Dashboards

5. System Design

Main Objects

Object	Purpose
Customer	Stores customer contact and address details
Shipment	Stores parcel, route, price and tracking details
Delivery Agent	Stores delivery staff information
Branch	Stores branch details
Delivery Update	Stores shipment status history
Invoice	Stores billing and payment details

Object Relationships

Child	Field	Parent	Relationship
Shipment	Customer	Customer	Lookup
Shipment	Assigned Agent	Delivery Agent	Lookup
Delivery Agent	Branch	Branch	Lookup
Delivery Update	Shipment	Shipment	Master-Detail
Invoice	Shipment	Shipment	Lookup

Basic data flow: Customer → Shipment → Delivery Agent → Delivery Update → Invoice → Reports.

6. Salesforce Setup

Step 1: Create Salesforce Developer Account

1. Open the Salesforce Developer signup page.
2. Enter your name, email, job title, college/company and country.
3. Submit the signup form.
4. Open the verification email and activate the account.
5. Set the password and security information.
6. Open the Salesforce Setup page.

Step 2: Open Setup

7. Log in to Salesforce.
8. Click the Gear icon.
9. Open Setup.
10. Use Quick Find to access Object Manager, Tabs, Flows, Profiles, Roles, Users, Reports and other required settings.

7. Create Custom Objects

Create the following six custom objects. The object names below are kept simple for project documentation; Salesforce will automatically add __c to custom object API names.

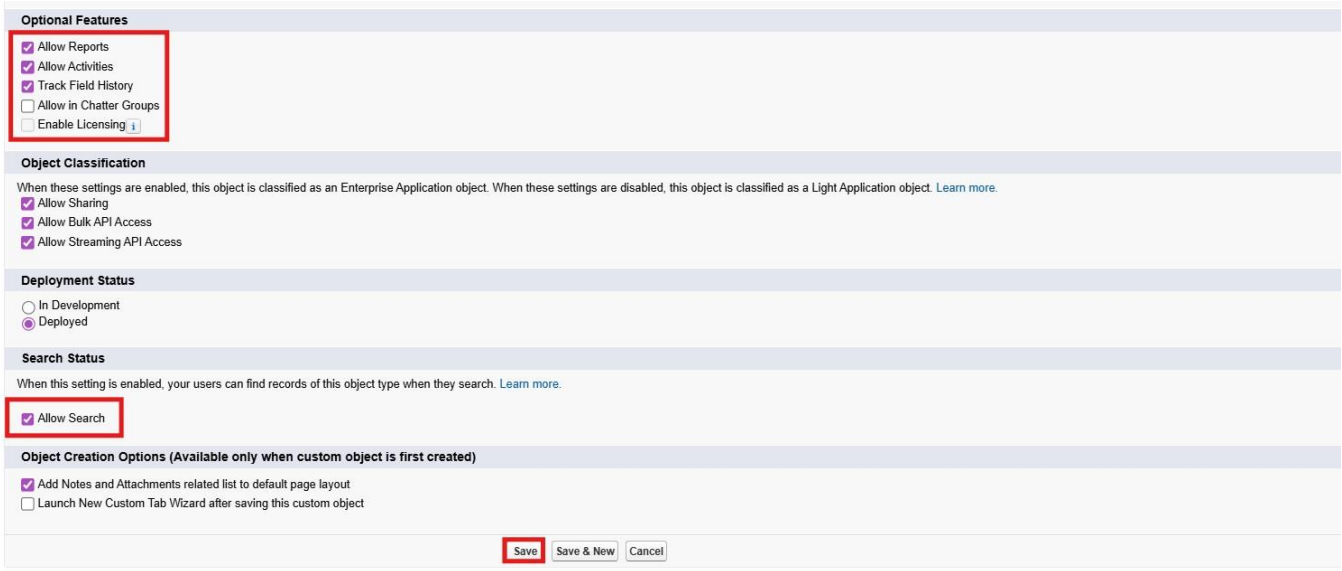
Object	Record Name	Purpose
Customer	Customer Name (Text)	Customer details
Shipment	Tracking Number (Auto Number)	Shipment details
Delivery Agent	Agent Name (Text)	Agent details
Branch	Branch Name (Text)	Branch information
Delivery Update	Update Number (Auto Number)	Status history
Invoice	Invoice Number (Auto Number)	Billing information

Common Steps to Create an Object

1. Go to Setup → Object Manager.
2. Select Create → Custom Object.
3. Enter the Label and Plural Label.
4. Choose the required Record Name type.
5. Enable Allow Reports where required.

Screenshot for Section 7 · Create Custom Objects

Related step: Common Steps to Create an Object ? enable the required options ? Save.



6.Save the object.

Suggested Auto Number Formats Shipment:

SHP-{00000}

Delivery Update: UPD-{0000}

Invoice: INV-{0000}

8. Create Fields Customer Fields

Field	Type	Required
Email	Email	Yes
Phone	Phone	Yes
Address	Text Area	Yes
Customer Type	Picklist: Regular, Business	Yes
City	Text	No
State	Text	No
Pincode	Text	No
Is Active	Checkbox	No

Shipment Fields

Field	Type	Required
Customer	Lookup → Customer	Yes
Source Address	Text Area	Yes
Destination Address	Text Area	Yes
Weight	Number	Yes
Price	Currency	Yes
Status	Picklist	Yes
Expected Date	Date	Yes
Assigned Agent	Lookup → Delivery Agent	No
Shipment Type	Picklist: Domestic, International	Yes

Delivery Agent Fields

Field	Type	Required
Email	Email	Yes
Phone	Phone	Yes
Branch	Lookup → Branch	Yes
Status	Picklist: Active, Inactive, On Leave	Yes
Joining Date	Date	No

Branch Fields

Field	Type	Required
Branch Code	Text	Yes
City	Text	Yes
State	Text	Yes
Manager Name	Text	No

Delivery Update Fields

Field	Type	Required
Shipment	Master-Detail → Shipment	Yes
Status	Picklist: In Transit, Out for Delivery, Delivered, Delayed	Yes
Update Date	Date/Time	Yes
Remarks	Text	No

Invoice Fields

Field	Type	Required
Shipment	Lookup → Shipment	Yes
Amount	Currency	Yes
Invoice Date	Date	Yes
	Picklist: Paid, Unpaid, Pending	

9. Create Relationships

Relationships connect the objects so that related information can be accessed from a record.

1. Open Setup → Object Manager → Shipment → Fields & Relationships → New.
2. Select Lookup Relationship and choose Customer. Label it Customer.
3. Create another Lookup Relationship from Shipment to Delivery Agent. Label it Assigned Agent.
4. Open Delivery Agent → Fields & Relationships → New → Lookup Relationship → Branch.
5. Open Delivery Update → Fields & Relationships → New → Master-Detail Relationship → Shipment.
6. Open Invoice → Fields & Relationships → New → Lookup Relationship → Shipment.
7. Save each relationship and verify it appears on the relevant object.

10. Create Tabs and Lightning App

Custom Tabs

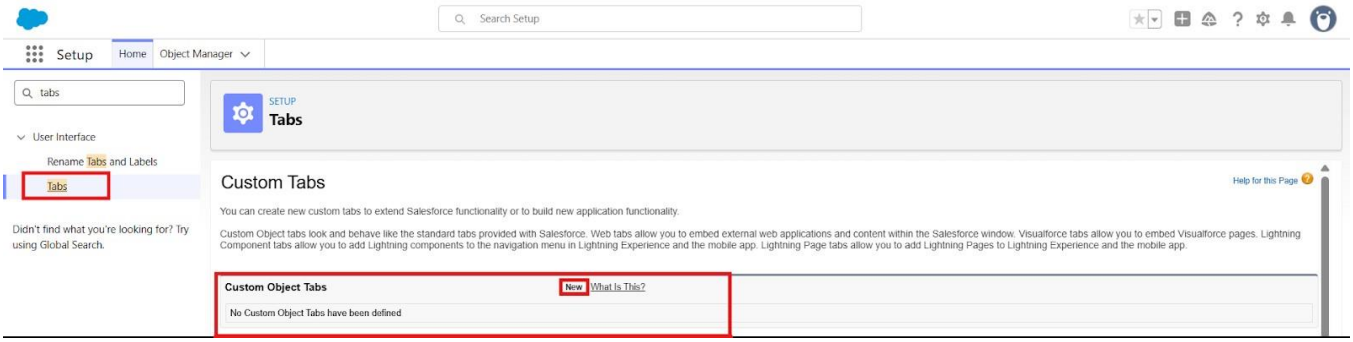
Create tabs for Customer, Shipment, Delivery Agent, Branch, Delivery Update and Invoice.

1. Go to Setup and search for Tabs.
2. Under Custom Object Tabs, click New.
3. Select the required object.
4. Choose a tab style.
5. Keep default settings unless a project requirement needs a change.
6. Click Next and Save.
7. Repeat for the remaining objects.

Create the Courier Management System App

Screenshot for Section 10 · Create Tabs

Related step: Setup ? Tabs ? Custom Object Tabs ? New.



1.Go to Setup → App Manager.

2.Click New Lightning App.

3.Enter the app name: Courier Management System.

4.Continue through the app setup screens.

5.Add Customer, Shipment, Delivery Agent, Branch, Delivery Update, 6. Invoice, Reports and Dashboards to the navigation items.

7.Add the required profile, such as System Administrator.

8.Save & Finish.

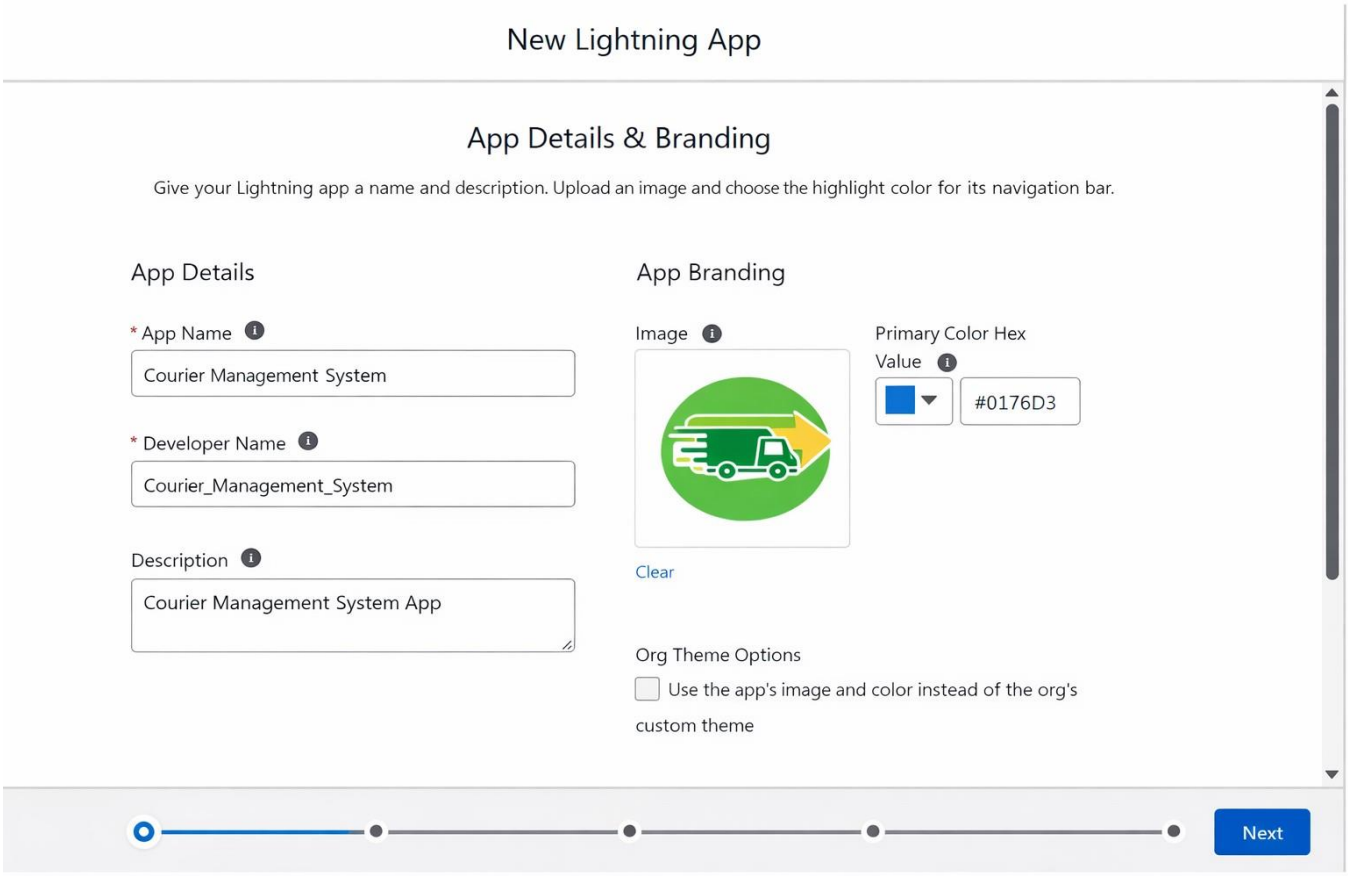
11. Validation Rules

Validation rules prevent incorrect records from being saved.

Object	Rule	Formula / Condition	Error
Customer	Email Required	ISBLANK(Email__c)	Email is required.
Customer	Phone Length	LEN(Phone__c) <> 10	Phone number must be 10 digits.
Customer	Pincode Length	LEN(Pincode__c) <> 6	Pincode must be 6 digits.
Shipment	Valid Weight	Weight__c <= 0	Weight must be greater than zero.

Screenshot for Section 10 · Create the Courier Management System App

Related step: Setup ? App Manager ? New Lightning App ? enter the app name.



Shipment	Valid Price	Price__c <= 0	Price must be greater than zero.
Shipment	Expected Date	Expected_Date__c < TODAY()	Expected delivery date cannot be in the past.
Delivery Agent	Valid Email	NOT(CONTAINS(Email__c, "@"))	Enter a valid email address.

Delayed Shipment Rule

For Delivery Update, require remarks when the status is Delayed.
Condition: AND(ISPICKVAL(Status__c, "Delayed"), ISBLANK(Remarks__c))
Error message: Remarks are required when status is Delayed.

Common Steps

1.Setup → Object Manager → select the object.

2.Open Validation Rules → New.

3.Enter Rule Name.

4.Enter the formula.

5.Click Check Syntax.

6.Enter the error message and location.

7.Save and activate the rule.

12. Page Layouts

Page layouts organize fields and related information so users can enter and view records easily.

Object	Suggested Sections
Customer	Customer Information; Address Details

C

Shipment	Shipment Details; Address Information; Delivery Information
Delivery Agent	Agent Details; Additional Information
Branch	Branch Information
Delivery Update	Update Information
Invoice	Invoice Details

1.Setup → Object Manager → select the object.

2.Open Page Layouts.

3.Open the default layout.

4.Add sections if required.

5.Place the related fields in the appropriate section.

13. Automation Using Flow

Flow: Automatically Update Shipment Status

Purpose: When a Delivery Update record is created, the related Shipment status should automatically change to the new status.

*Go to Setup → Flows → New Flow.

*Select Record-Triggered Flow and click Create.

*Select Delivery_Update__c as the object.

*Set the trigger to run when a record is created.

*Leave the condition as None so the flow can run for every new delivery update.

*Choose Actions and Related Records.

*Add an Update Records element.

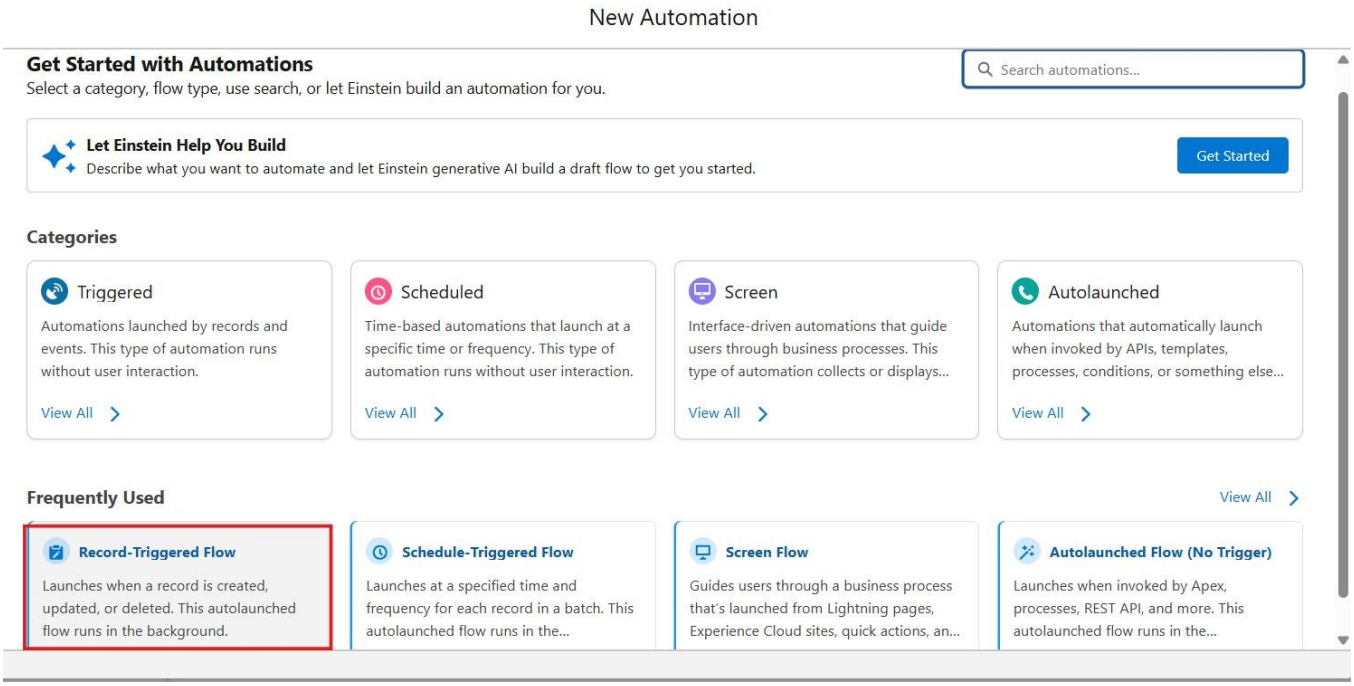
*Select Shipment__c as the record to update.

*Match Shipment Id with the Shipment value from the new Delivery Update.

*Set Shipment Status__c equal to the Delivery Update Status__c.

Screenshot for Section 13 · Automation Using Flow

Related step: Setup ? Flows ? New Flow ? select Record-Triggered Flow.



*Save the flow as Auto_Update_Shipment_Status.

*Activate the flow.

Expected Result

Creating a Delivery Update with a new status should automatically update the related Shipment record.

14. Profiles and Permissions

Profiles control what users can access. For this project, separate profiles can be used for different responsibilities.

Profile	Main Purpose	Key Access
Courier Admin	Full system administration	All project objects and setup
Branch Manager	Manage branch operations	Customer, Shipment, Agent, Branch, Updates, Reports
Delivery Agent	Update deliveries	Shipment and Delivery Update; limited access
Customer Support	Handle customer tracking	Customer, Shipment, Delivery Update and Invoice read access

Create a Custom Profile Setup

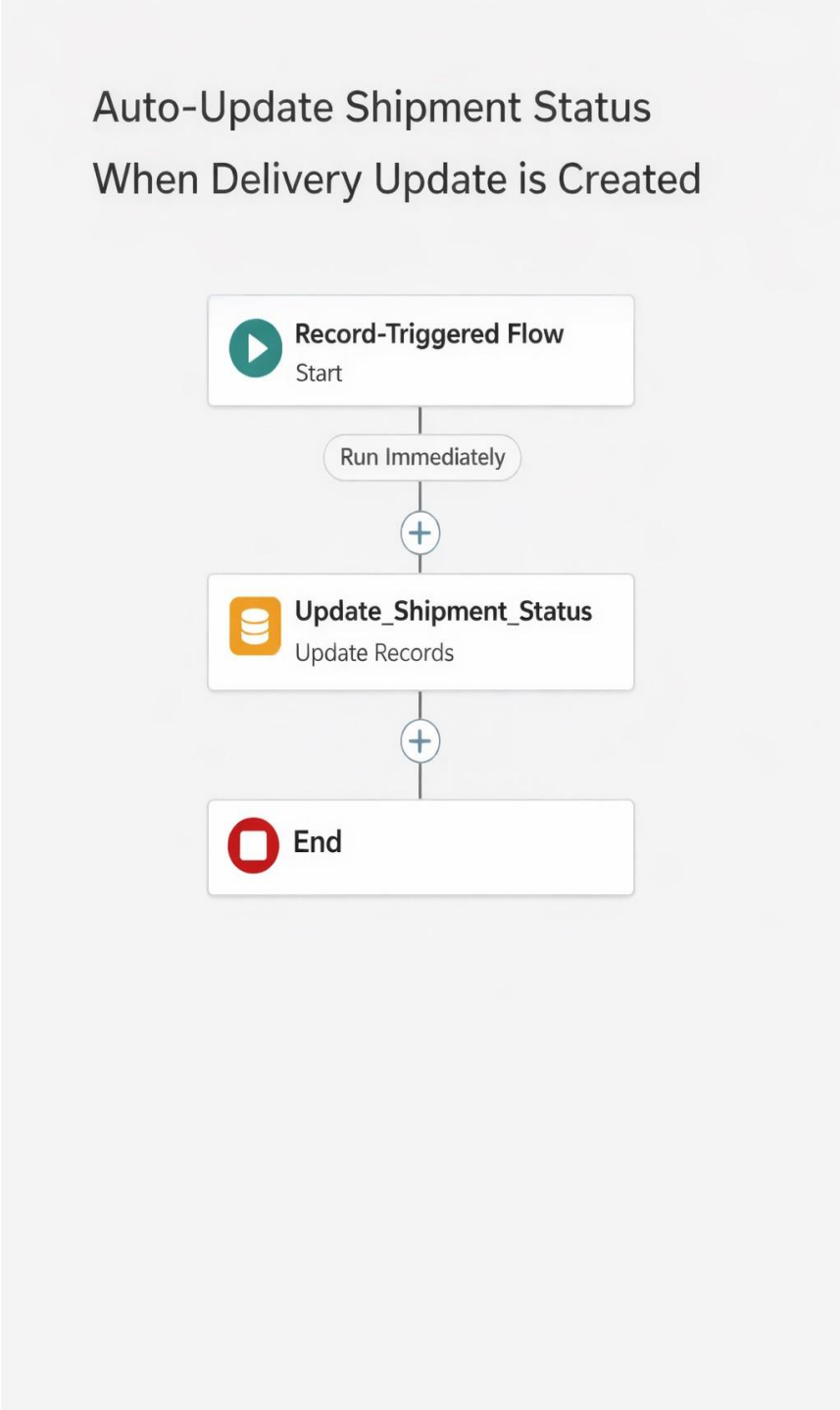
→ Profiles.

Click New Profile.

Select an appropriate existing profile as the base.

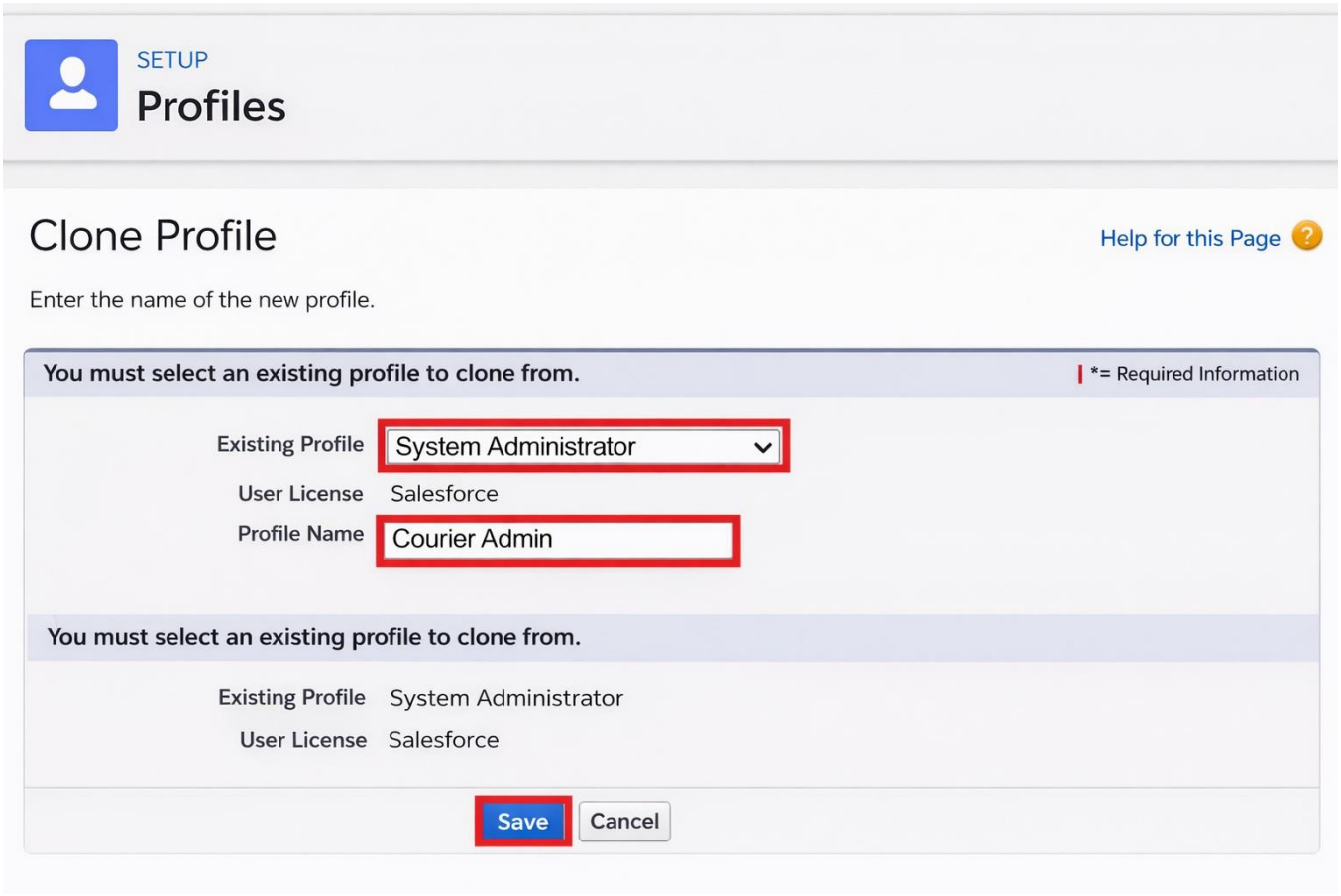
Screenshot for Section 13 · Automation Using Flow

Related step: Add an Update Records element to update the related Shipment status.



Screenshot for Section 14 · Profiles and Permissions

Related step: Setup ? Profiles ? New Profile / clone an existing profile.



Enter the required profile name.

Save.

Edit object, tab and app permissions according to the user's role.

15. Roles and Users

Role Hierarchy

Level	Role	Purpose
1	Courier Admin	Overall administration
2	Branch Manager	Branch-level management
3	Delivery Agent	Delivery operations
4	Customer Support	Customer and tracking support

Create Roles

Setup → Roles → Set Up Roles.

Add Courier Admin under the top-level role. Add

Branch Manager under Courier Admin.

Add Delivery Agent under Branch Manager.

Add Customer Support under Branch Manager. Save

each role.

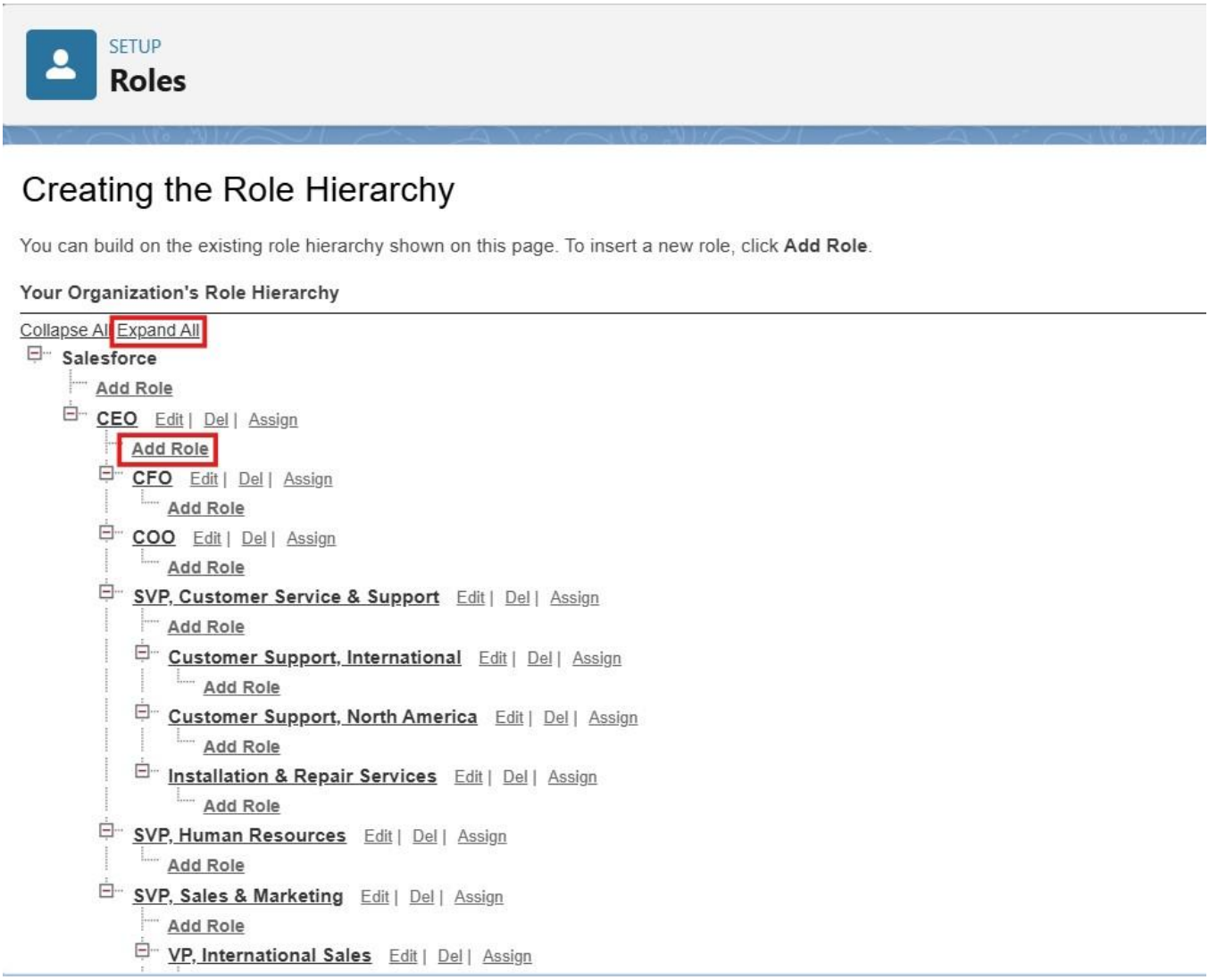
Create Users

1. Setup → Users → New User.
2. Enter first name, last name, email, username and alias. 62.

   Select the correct role.
3. Select the appropriate user license available in the org.
4. Select the matching profile.
5. Save the user.

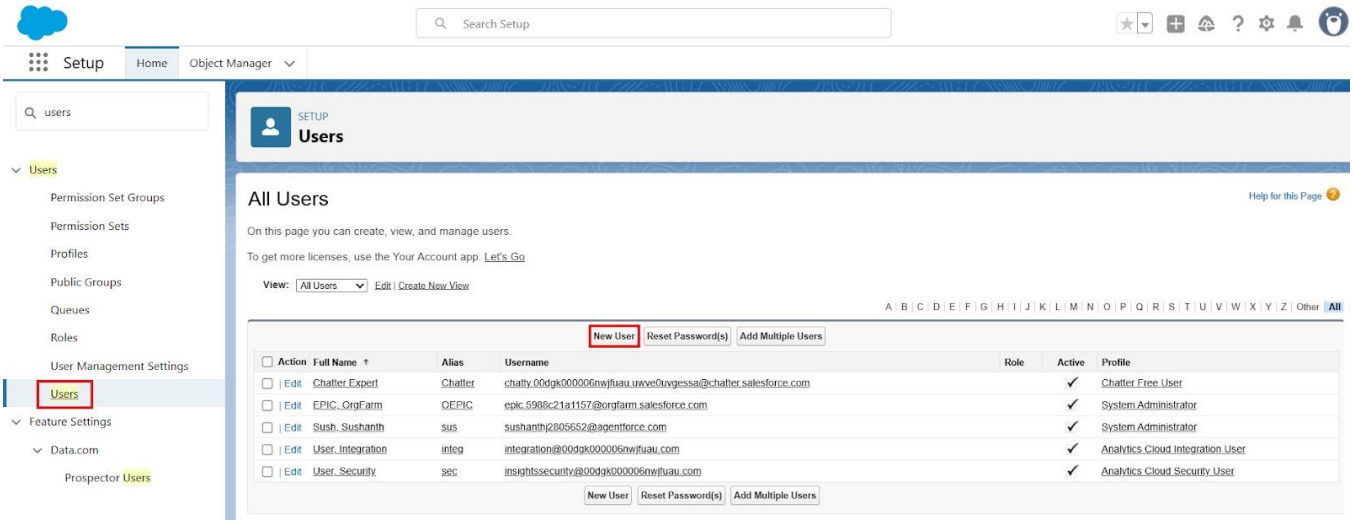
Screenshot for Section 15 · Roles and Users

Related step: Setup ? Roles ? Set Up Roles ? create the role hierarchy.



Screenshot for Section 15 · Create Users

Related step: Setup ? Users ? New User.



16. Reports and Dashboard

Recommended Reports

Report	Main Fields	Purpose
Shipment Status Summary	Tracking Number, Customer, Status, Expected Date, Assigned Agent	Track shipment progress
Agent Performance	Agent, Branch, Shipment count, Status	Review agent activity
Revenue by Branch	Tracking Number, Customer, Price, Invoice Status, Branch	Review revenue information

Create a Report

1. Open App Launcher → Reports.
2. Click New Report.
3. Choose the relevant report type.
4. Click Continue.
5. Add the required columns.
6. Add filters such as This Month when needed.
7. Group the records by Status, Agent or Branch as required.
8. Save & Run the report.

Create Dashboard

o Open App Launcher → Dashboards. o Click New Dashboard. o Enter the name Courier Management Dashboard. o Choose the required folder. o Add a shipment status component using a donut chart. o Add an agent performance component using a bar chart. o Add a revenue component using a column chart. o Save the dashboard.

The dashboard provides a quick visual summary of shipment status, agent activity and revenue.

17. Functional Testing

Testing is performed after the main configuration is completed.

Test Case	Action	Expected Result
TC-01	Create a Customer with valid details	Customer record is saved
TC-02	Leave required email blank	Validation error is displayed
TC-03	Enter an invalid phone length	Phone validation prevents saving
TC-04	Create a Shipment with valid values	Shipment is created with a tracking number
TC-05	Enter zero/negative weight	Weight validation prevents saving
TC-06	Enter a past expected date	Date validation prevents saving
TC-07	Create a Delivery Update	Related Shipment status is updated by Flow
TC-08	Create a delayed update	Validation error is displayed
	without remarks	
TC-09	Run Shipment Status report	Shipment records appear in the report
TC-10	Open Dashboard	Charts display project data

Testing Evidence

For the final submission, capture screenshots from the student's own Salesforce org for important completed stages such as objects, records, validation errors, Flow activation, users, reports and dashboard. Insert only relevant screenshots.

18. Advantages and Limitations

Advantages

Centralizes courier information in one platform.

Reduces manual data entry errors through validation.

Automates shipment status updates.

Supports role-based access.

Provides reports and dashboards for monitoring.

Limitations

The system depends on the Salesforce platform.

Initial configuration requires time and proper setup.

Advanced integrations may require additional Salesforce features or services.

19. Future Scope

GPS-based live shipment tracking.

SMS or email delivery notifications.

Mobile application for customers and delivery agents.

Integration with payment services.

AI-based route or delivery optimization.

Expanded analytics for branch and delivery performance.

20. Conclusion

The Courier Management System provides a centralized approach to managing courier operations using Salesforce. The project covers custom objects, fields, relationships, validation rules, page layouts, Flow automation, user security, reports and dashboards. The completed system can help organize shipment information, improve tracking
visibility, reduce data errors and provide useful operational information to managers and staff.

Appendix: Quick Implementation Checklist

1.Salesforce Developer Org created and verified

2.Customer object created

3.Shipment object created

4.Delivery Agent object created

5.Branch object created

6.Delivery Update object created

7.Invoice object created

8.Required fields created

9.Object relationships created

10.Custom tabs created

11.Courier Management System Lightning App created

12.Validation rules tested 13.Page layouts arranged

14.Shipment status Flow created and activated

15.Profiles configured

16.Roles created

17.Users created

18.Reports created

19.Dashboard created

20.Functional testing completed

— End of Project Document —